

Fall Semester 2026

Course Syllabus, version 23.06.2026

GRAD-E1393: Development Economics

1. General information

Course Format	Onsite
Instructor(s)	Arianna Ornaghi
Instructor's E-mail	ornaghi@hertie-school.org
Instructor's Office	2.41.1
Assistant	Andrea Derichs-Carlin, derichs@hertie-school.org
Instructor's Office Hours	TBD

Link to [Study, Examination and Admission Rules and MIA, MDS and MPP Module Handbooks](#)

For information on **course room, times and session dates**, please consult the [Course Schedule](#) on *MyStudies*.

Instructor Information:

Arianna Ornaghi is an Assistant Professor of Economics at the Hertie School. Arianna is an applied microeconomist who specializes in empirical political economy. Before joining the Hertie School, Arianna was a British Academy Postdoctoral Research Fellow at the University of Warwick. She earned a PhD in Economics from the Massachusetts Institute of Technology in 2017 and holds BSc and MSc degrees in Economics and Social Sciences from Bocconi University.

2. Course Contents and Learning Objectives

Course contents:

In recent decades, social protection programs have become increasingly common in low- and middle-income countries. This course discusses the challenges that governments in these countries face when designing such programs and the taxation systems needed to finance them. Central to the course is a methodological focus on randomized control trials: how they can be designed to evaluate policies and understand economic mechanisms, how different implementation challenges can be addressed, and how to analyze and interpret their results.

We will begin by considering the specificities of taxation systems in developing countries, discussing the role played by instruments such as the value added tax and tax administration. We will then examine how social protection programs can be structured to address the challenges associated with large informal

sectors and weak state capacity: from how to target recipients, to what type of transfers are appropriate, to how to deal with governance issues. We will also consider the general equilibrium effects of social protection schemes. The course might further explore thematic extensions including insurance, migration, and climate change adaptation.

Learning objectives:

By the end of the course, students will be able to:

- Describe the key issues concerning taxation systems and social protection programs in developing countries and discuss what state-of-the-art empirical research in economics says about them;
- Understand in detail methodological issues related to using randomized controlled trials to evaluate policies;
- Approach advanced research papers in economics and distill key messages and implications.

Target group:

2nd year MPP, MIA, and MDS students with a particular interest in development economics and randomized control trials.

Teaching style:

The course will mostly follow a seminar format. There will be short presentations of research papers by students or the professor, followed by an in-depth class discussion that will focus on methodological and policy issues. The sections that will introduce the course and as well the ones about RCT design and power calculations will be an exception to this and follow a more traditional lecture format.

Prerequisites:

This is an applied microeconomics course, where students are required to read state-of-the-art papers in economics. Two sets of skills are important to be successful in this course.

First, it is important to have a basic understanding of microeconomic concepts as taught in the Economics I course. This is to be able to understand the general conceptual framework of the readings that we will be discussing (although we will generally not devote much time to the discussion of theoretical models).

Second, it is fundamental to have a good understanding of basic statistics and regression analysis (e.g., being able to interpret a regression coefficient). This means that students should be comfortable with the material of the Statistics I course (or a close equivalent). In addition to this, it is recommended that students have a basic understanding of the problem of causal identification and how randomized control trials can help achieve causality. Having taken Statistics II is recommended.

Basic knowledge of R or Stata will be needed to perform power calculations.

Diversity Statement:

I am committed to creating an inclusive classroom, in which a wide range of perspectives are valued and heard. Being this a course about development economics, the reading list naturally spans a number of different countries in Africa, Asia, and Latin America. To provide an overview of the geographic scope of the course, I have listed the country that each paper refers to with the required readings below. However, it is important to acknowledge that most of the papers included in the reading list are authored by researchers from universities in the U.S. and Europe.

3. Grading and Assignments

The grade for this course comes from three main components: 1) class participation; 2) in class presentation; 3) an RCT design paper. The RCT design paper is split into different assignments, to be handed in at different times during the course as detailed below, so that students can receive feedback along the way and incorporate it in the final writeup.

Composition of Final Grade:

Assignment 1: Presentation.	Deadline: day of the presentation.	Submit via Moodle.	15%
Assignment 2: Setting, intervention, and conceptual framework.	Deadline: October 11 th 2026.	Submit via Moodle.	20%
Assignment 3: Study design, data, and power calculations.	Deadline: November 15 th 2026.	Submit via Moodle.	35%
Assignment 4: Complete design.	Deadline: December 11 th 2026.	Submit via Moodle.	20%
Participation grade			10%

Assignment Details

Assignment 1: In-class presentation (20%)

Students will prepare and deliver a short presentation of one of the required readings for the course. The presentation should introduce the topic, outline the research question, discuss the data/methodology, and describe the main results. Successful presentations will identify the key elements of the paper independently and explain them clearly. Students might be asked to complete this assignment independently or in small groups, depending on the size of the class. More details on the assignment will be provided at the beginning of the course.

Assignment 2: Setting, intervention, and conceptual framework (group work, 20%)

In small groups, the students will design a randomized control trial to evaluate a policy of choice. In this assignment, students are asked to set the stage for the RCT: what is the policy problem you aim to

address, what is the solution that you propose to evaluate (in other words, what is your intervention), and what conceptual framework justifies your approach? More details on the assignment will be provided in due time.

Assignment 3: Study design, data, and power calculations (group work, 35%)

In the same groups, the students will further develop the RCT design by describing the treatment arms, data and measurement, and implementing power calculations to determine the sample size needed to evaluate the intervention. In addition, each group will prepare a poster summarizing their RCT design, to be presented during a poster session in the last class meeting on November 24th. During the session, groups will present and defend their design to peers and the instructor, and engage critically with other groups' work. More details on the assignment will be provided in due time.

Assignment 4: Complete design (group work, 20%)

This assignment finalizes the RCT design. Incorporating feedback from the previous two assignments, this assignment will present a complete RCT design. Additional issues to be addressed in the final paper will include RCT challenges (e.g., attrition, etc.) and potential ethical considerations. The final design will be presented to the class during a poster session in the last session of the course. More details on the assignment will be provided in due time.

Participation grade (10%)

The participation grade rewards active, consistent, and high-quality engagement in class discussions across all sessions. Students are expected to come to every session having read the required papers and prepared to contribute. The instructor may cold call students to ask questions about the papers; this is intended to support engagement rather than to penalize, and a thoughtful response that acknowledges uncertainty or confusion is entirely acceptable.

Late submission of assignments: For each day the assignment is turned in late, the grade will be reduced by 10% (e.g. submission two days after the deadline would result in 20% grade deduction).

Attendance: Students are expected to be present and prepared for each class session. Active participation during lectures and seminar discussions is essential. Please note that students can miss up to two sessions (out of twelve) if no course assignments are affected. **Students do not need to inform the course instructor(s) in advance if they are unable to attend a session.** For further information please consult the [Examination Rules](#) §10.

Academic Integrity: The Hertie School is committed to the standards of good academic and ethical conduct. Any violation of these standards shall be subject to disciplinary action. Plagiarism, misuse of AI, free riding in group work, and other deceitful actions are not tolerated. See [Examination Rules](#) §16, the Hertie [Plagiarism Policy](#), and [the Hertie Guidelines for Artificial Intelligence Tools](#).

Compensation for Disadvantages: If a student furnishes evidence that they are not able to take an examination as required in whole or in part due to disability or permanent illness, the Examination Committee may upon written request approve learning accommodation(s). In this respect, the submission of adequate certificates may be required. See [Examination Rules](#) §14.

Extenuating circumstances: An extension can be granted due to extenuating circumstances (i.e., for reasons like illness, personal loss or hardship, or caring duties). In such cases, please contact the course instructor and Examination Office *in advance* of the assignment deadline.

Policy on the use of AI tools for assignments

The use of generative AI tools (such as ChatGPT) to complete assignment is permitted within the limit that will be specified for each assignment. Students must adhere to the specific instructions provided for each assignment. Where AI tools are used, students are expected to clearly indicate how they were used and to ensure the originality and academic integrity of their work. **Students must include a Statement of Authorship in all written assignments.** Improper or undisclosed use of AI tools may be considered a violation of academic integrity policies. Please consult the course instructor in case of doubt. Further guidance is available in the [Artificial Intelligence Tools at the Hertie School - Teaching Guidelines for Instructors and Students](#), including a template for the **Statement of Authorship**, under section 1.1.

4. General Readings

For students who are interested in updating their knowledge of empirical policy evaluation, I recommend the following textbook (Chapters 1, 3, and 4 in particular):

Gertler, Paul J. et al. Impact Evaluation in Practice (2nd Ed.), Washington: The World Bank. 2016. [available for free download [here](#)].

The following book provides is a great reference for anyone interested in designing an RCT:

Glennerster, R. and Takavarasha, K., 2014. Running Randomized Evaluations: A Practical Guide. [available as an e-book [here](#)].

5. Course Sessions and Readings

All course readings can be accessed on the course Moodle page.

Please note that fall semester Moodle course pages are available until 31 March of the following semester. Spring semester Moodle courses are available until 31 September of the following semester. After that, in accordance with German copyright law, they cannot be accessed.

Please note that this is a preliminary reading list, and there may be a handful of updates to the provided references before the start of the course. The updated syllabus will be provided on Moodle in due time.

Session 1: Introduction to the course

Required Readings

There are no required readings for this session, but I recommend skimming through the following papers if you are interested in an overview of the topics covered in the course:

Banerjee, A., Hanna, R., Olken, B.A. and Lisker, D.S., 2024. Social protection in the developing world. *Journal of Economic Literature*, 62(4), pp.1349-1421.

VoxDevLit: Taxation and Development. Version 1.

Session 2: Randomized Control Trials

Required Readings	Gibson, M. and Sautmann, A., 2023. Introduction to Randomized Evaluations. https://www.povertyactionlab.org/resource/introduction-randomized-evaluations Deaton, A., 2019. Randomization in The Tropics Revisited: A Theme and Variations. https://www.theindiaforum.in/article/randomization-tropics-revisited
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Session 3: Tax systems

Required Readings	Bachas, P., Gadenne, L. and Jensen, A., 2024. Informality, Consumption Taxes, and Redistribution. <i>Review of Economic Studies</i>, 91(5), pp.2604-2634. [Cross-country] Pomeranz, D., 2015. No Taxation without Information: Deterrence and Self-Enforcement in the Value Added Tax. <i>American Economic Review</i>, 105(8), pp.2539-69. [Chile]
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Session 4: Tax administration

Required Readings	Bergeron, A., Tourek, G. and Weigel, J., 2024. The State Capacity Ceiling on Tax Rates: Evidence from Randomized Tax Abatements in the DRC. <i>Econometrica</i>, 92(4), 1163-1193. [DRC] Dzansi, J., Jensen, A., Lagakos, D. and Telli, H., 2025. Technology and Tax Capacity: Evidence from Local Governments in Ghana. Working paper. [Ghana]
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Session 5: Program design

Required Readings	Balboni, C., Bandiera, O., Burgess, R., Ghatak, M. and Heil, A., 2022. Why Do People Stay Poor?. <i>The Quarterly Journal of Economics</i>, 137(2), pp.785-844. [Bangladesh] Haushofer, J. and Shapiro, J., 2016. The Short-Term Impact of Unconditional Cash Transfers to the Poor: Experimental Evidence from Kenya. <i>The Quarterly Journal of Economics</i>, 131(4), pp.1973-2042. [Kenya]
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Session 6: Targeting

Required Readings	Alatas, V., Banerjee, A., Hanna, R., Olken, B.A., Purnamasari, R., and Wai-Poi, M., 2016. Self-targeting: Evidence from a Field Experiment in Indonesia. <i>Journal of Political Economy</i>, 124(2), pp.371-427. [Indonesia] Alatas, V., Banerjee, A., Hanna, R., Olken, B.A. and Tobias, J., 2012. Targeting the Poor: Evidence from a Field Experiment in Indonesia. <i>American Economic Review</i>, 102(4), pp.1206-40. [Indonesia]
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Session 7: Governance

Required Readings	Banerjee, A., Hanna, R., Kyle, J., Olken, B.A. and Sumarto, S., 2018. Tangible Information and Citizen Empowerment: Identification Cards and Food Subsidy Programs in Indonesia. <i>Journal of Political Economy</i>, 126(2), pp.451-491. [Indonesia] Muralidharan, K., Niehaus, P. and Sukhtankar, S., 2016. Building State Capacity: Evidence from Biometric Smartcards in India. <i>American Economic Review</i>, 106(10), pp.2895-2929. [India]
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Session 8: General equilibrium

Required Readings	Egger, D., Haushofer, J., Miguel, E., Niehaus, P. and Walker, M.W., 2022. General Equilibrium Effects of Cash Transfers: Experimental Evidence from Kenya. <i>Econometrica</i>, 90(6), pp. 2603-2643. [Kenya] Muralidharan, K., Niehaus, P. and Sukhtankar, S., 2023. General Equilibrium Effects of (Improving) Public Employment Programs: Experimental Evidence from India. <i>Econometrica</i>, 91(4), pp. 1261-1295. [India]
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Session 9: Power calculations

Required Readings	Glennerster, R. and Takavarasha, K., 2014. Running Randomized Evaluations: A Practical Guide. Sections 6.1 to 6.3 (power calculations).
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Session 10: Insurance

Required Readings	Banerjee, A., Finkelstein, A., Hanna, R., Olken, B.A., Ornaghi, A. and Sumarto, S., 2021. The Challenges of Universal Health Insurance in Developing Countries: Experimental Evidence from Indonesia's National Health Insurance. <i>American Economic Review</i>, 111(9), pp.3035-63. [Indonesia] Cai, J., De Janvry, A. and Sadoulet, E., 2020. Subsidy Policies and Insurance Demand. <i>American Economic Review</i>, 110(8), pp.2422-2453. [China]
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Session 11: Climate change adaptation

Required Readings	Lane, G., 2024. Adapting to Climate Risk with Guaranteed Credit: Evidence from Bangladesh. <i>Econometrica</i>, 92(2), pp.355-386. [Bangladesh] Patel, D., 2025. Environmental Beliefs and Adaptation to Climate Change. Working Paper. [Bangladesh]
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Session 12: Poster Session

Required Readings

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